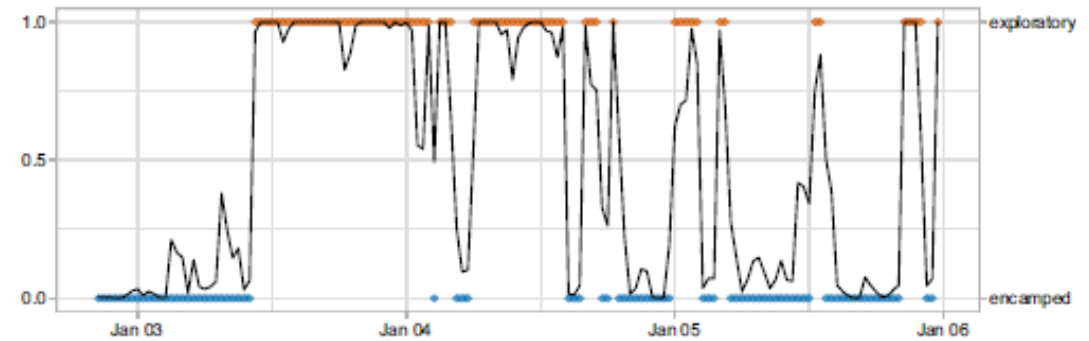


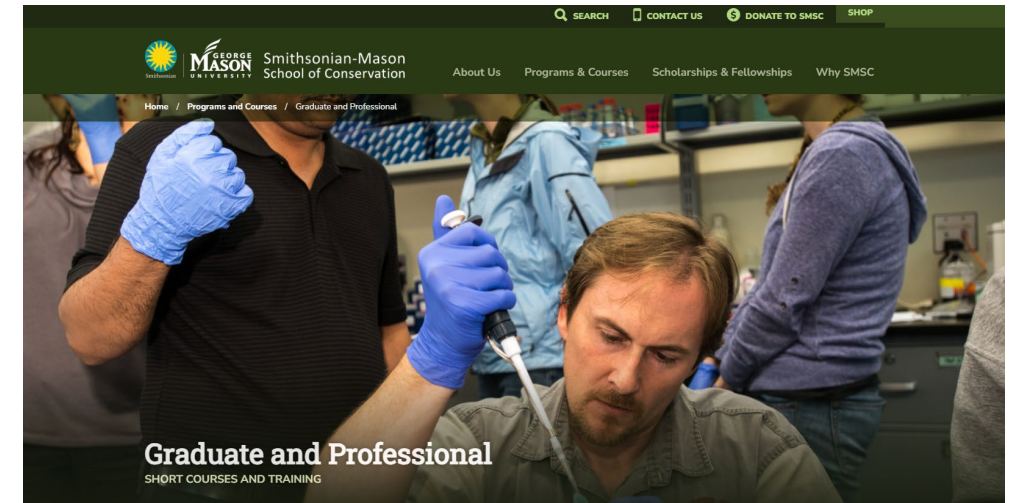
Introduction to Animal Movement Analyses

Categorizing Behavioral States

$$\begin{matrix} & s_1 & s_2 & s_3 \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.05 & 0.3 & 0.65 \end{pmatrix} \end{matrix}$$



This lecture is based on a graduate training course taught at the Smithsonian-Mason School of Conservation. Many thanks to Dr. Joe Kolowski for sharing his notes.

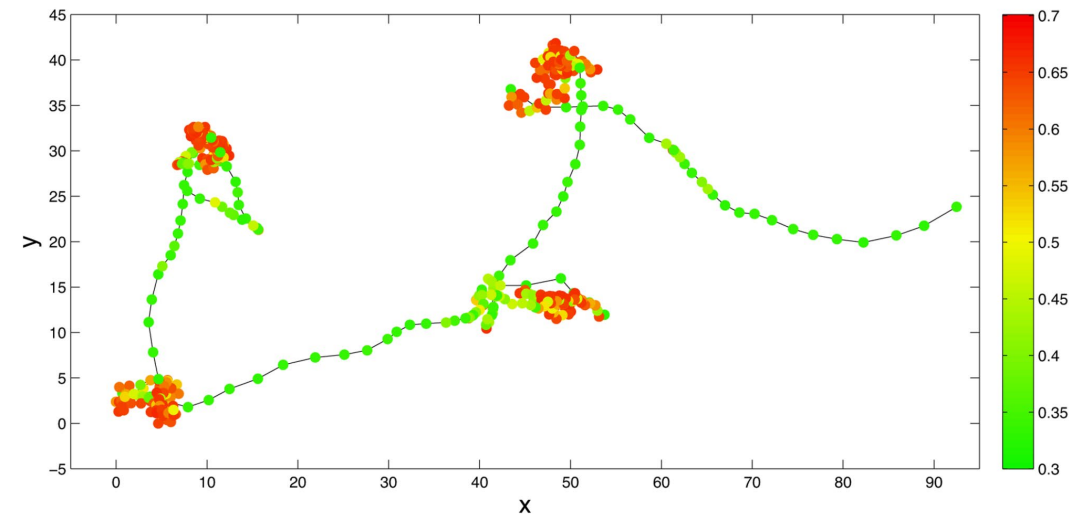


<https://smconservation.gmu.edu/programs/graduate-and-professional/>

Behavioral Partitioning

- Hidden Markov Models (State-switching State Space Models)
- Behavioral Change Point Analysis (BCPA)
- First Passage Time/Residence Time
- Multi-state Random Walk

See Gurarie et al. 2016
for summary and comparison of methods



Applications of Behavior State Assignment

Exploratory
(What is the animal doing?)

- Metrics to describe and quantify the movement track
 - Are there distinct behaviors? If so, how many?
 - What distinguishes the behaviors?
 - When and where do the behaviors occur?

Explanatory
(Why is the animal doing it?)

- Associate behaviors and transitions with covariates
 - Season
 - Environmental cues
 - Predation or conspecifics

Predictive
(Can we anticipate movement from behavior?)

- If we modify the environment in some way, how will movement behavior change?

Hidden Markov Models

- Discrete time, discrete state, state space models
- Operate on **time-series data** for step lengths and turning angles
 - Could also be used on accelerometry data, but also occupancy data, capture-recapture data...
- Parameters of step length distribution and turning angle distribution determined by an underlying observed state

Hidden Markov Models

- Observation process and latent state process
 - Locations are the observation process, behavior state is the latent state process
 - Latent state process taken to be a Markov Chain
 - Discrete random variable, $(1, 2, \dots, N)$
 - N is # of potential states (typically ≤ 4)
 - Data distributions (step length and turn angle) vary by state
 - Transition probability matrix
 - Governs the sequence of switching most likely and how probable these switches are

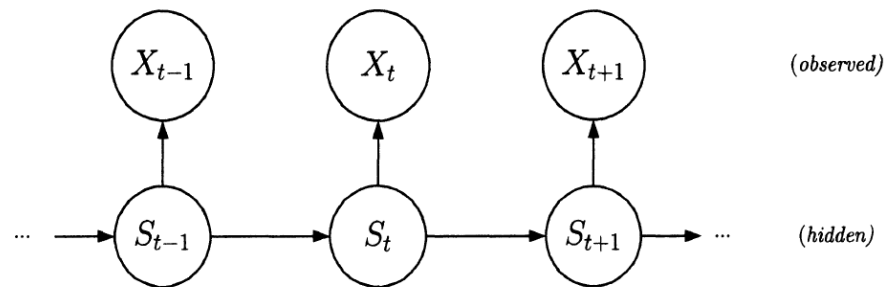


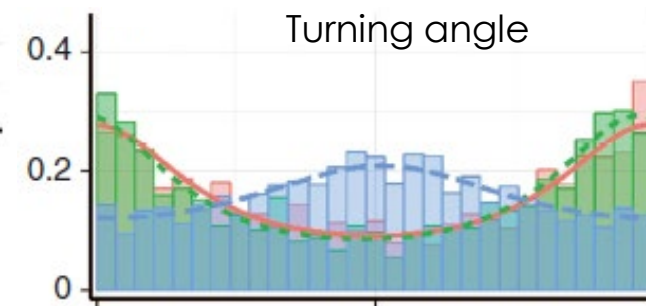
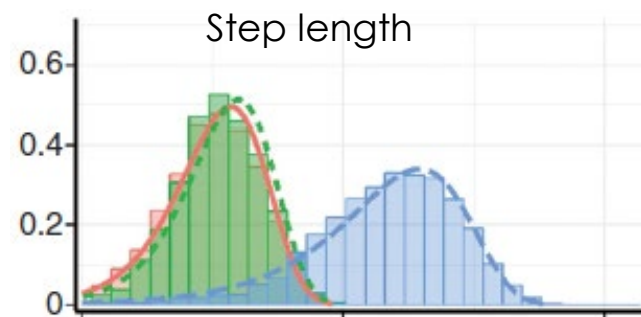
Figure 1. Dependence structure of an HMM in its most basic form.

Pohle et al. 2017

Implementing HMMs in R

- Models estimate:
 - 1) Transition probabilities
 - 2) Mean and variance of step length and turn angle for all states
- Transition probabilities can be modeled as function of covariates

$$\begin{matrix} & s_1 & s_2 & s_3 \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.05 & 0.3 & 0.65 \end{pmatrix} \end{matrix}$$



HMM Parameters for Animal Movement

- $N = \#$ of states
- Distribution of step lengths (for each state)
 - Gamma, Weibull, log-normal and exponential distributions
 - Mean
 - Variance (SD)
- Distribution of turning angles (for each state)
 - von Mises and wrapped Cauchy distributions
 - Mean
 - Clustering parameter

$$\begin{matrix} & s_1 & s_2 & s_3 \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.05 & 0.3 & 0.65 \end{pmatrix} \end{matrix}$$



Implementing HMMs in R

- Connecting location data to latent states (decoding algorithms)
 - Viterbi
 - Finds most probable sequence of states that generated the data
 - Stationary probabilities available
- Assume regular sampling with negligible error
 - Paths can be smoothed prior to analysis if error on same scale as step lengths
 - No date/time information is included
- User specifies # of state (≤ 4) and models with varying states can be compared with AIC



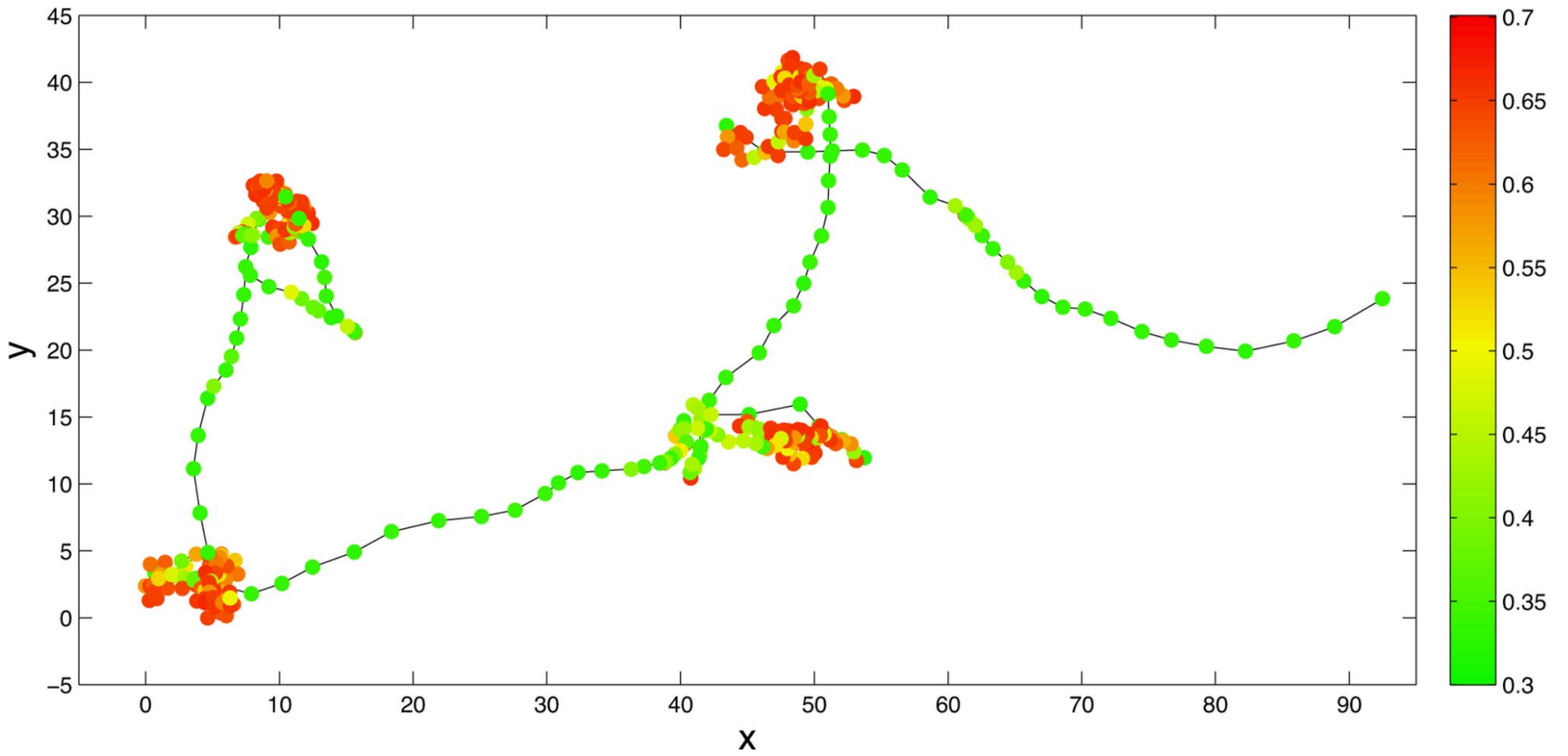
Hidden Markov Models with momentuHMM

```
my_mod <- fitHMM(data = my_move_object,  
  nbStates= 2, # Needs to be specified ( $\leq 4$ )  
  dist = list(step = "gamma", angle = "vm"), # Distributions  
  Par0 = list(step = stepPar0, angle = anglePar0), # Starting values  
  formula = ~ 1)
```

```
plot(my_mod)
```

Ability to investigate impact of covariates on transition probabilities





Postlethwaite and Dennis 2013

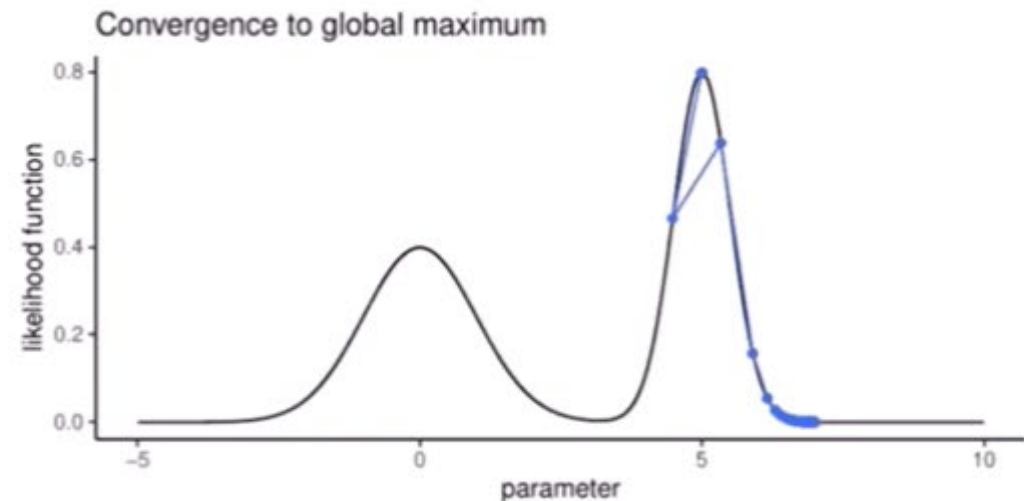
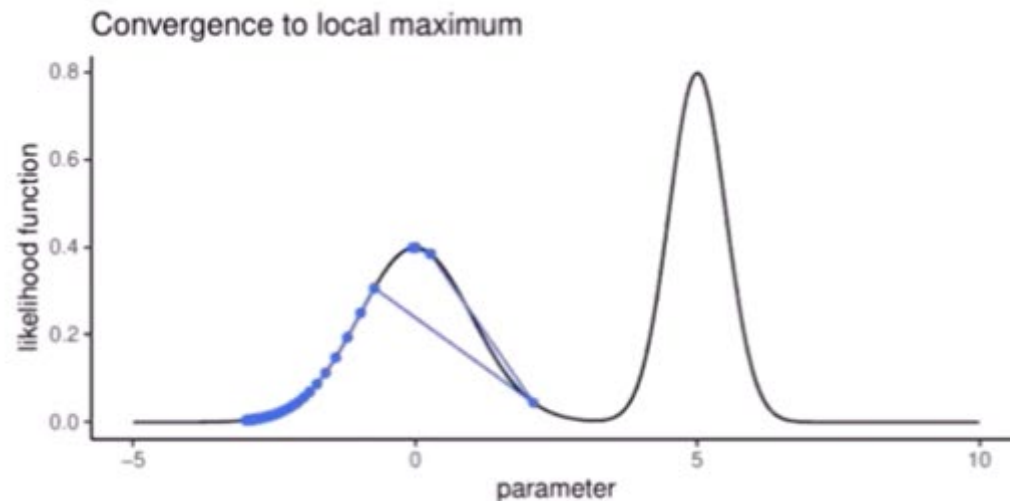


Common Challenges

Choosing Starting Values

HMMs are fitted by numerical optimization (to find parameters that maximise the likelihood)

- User needs to set a starting point
- Convergence to “global” maximum (i.e., best model) requires a good starting point



Partial solution: try many starting values; look at data to pick plausible values

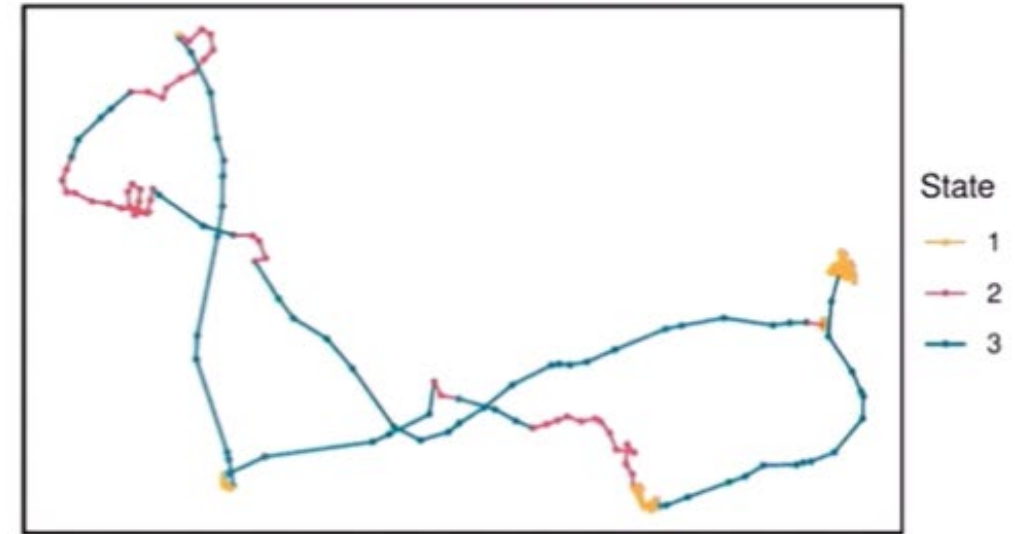
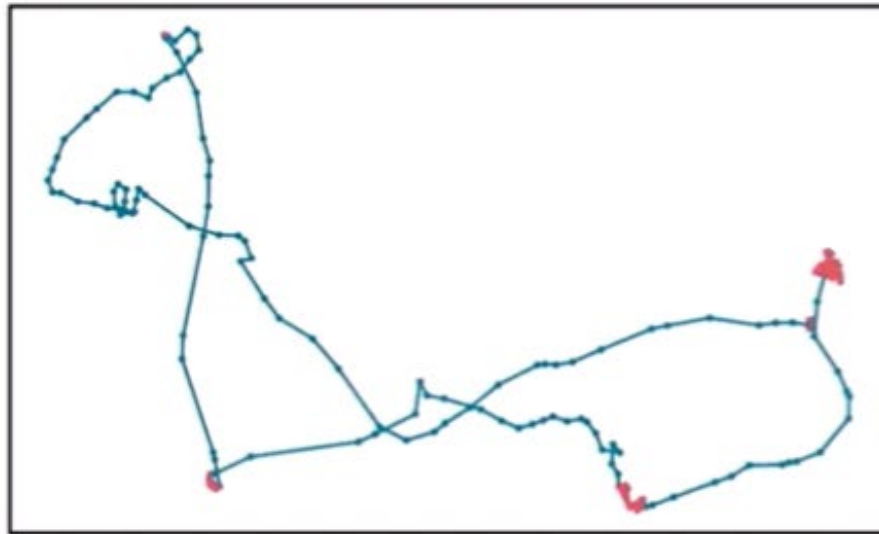
Slide from T. Michelot

Common Challenges

Choosing the Number of States

We need to choose the number of states

- Most studies use 2 states (“slow” and “fast”)
- 3-4 can sometimes be interpreted biologically – beyond that it may be difficult
- AIC/BIC often select much more complex models



Partial solution: Use a combination of model checking and biological knowledge

Slide from T. Michelot

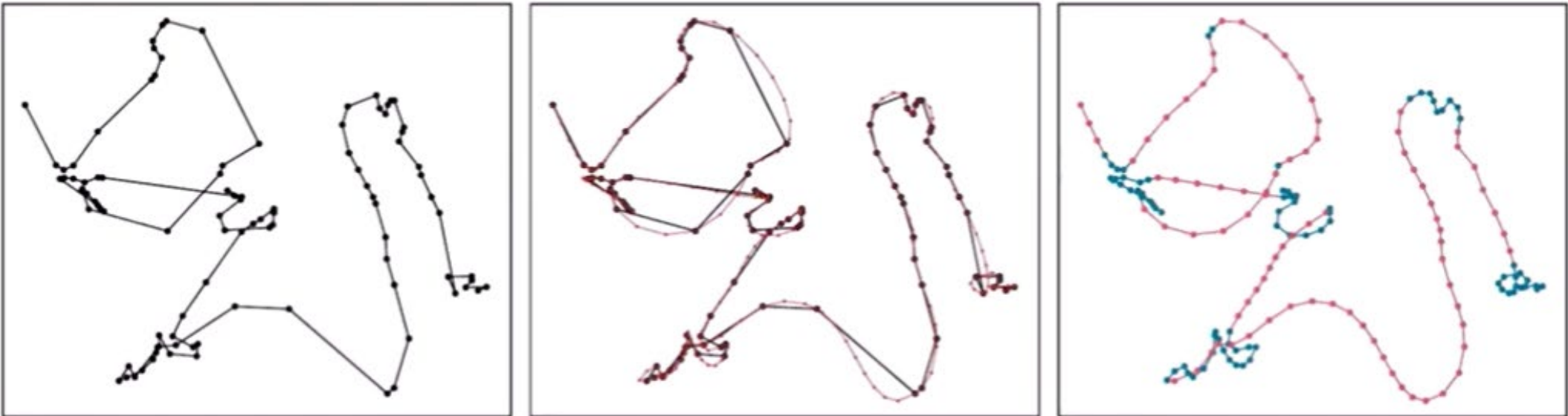


Common Challenges

Irregular or Noisy Locations

HMM requirements:

- Measurement error should be negligible
- Locations should be observed at regularly time intervals



Partial solution: Filter noise and regularize tracks with a state-space or simulation approach

Slide from T. Michelot

Multiple Imputation Procedure:

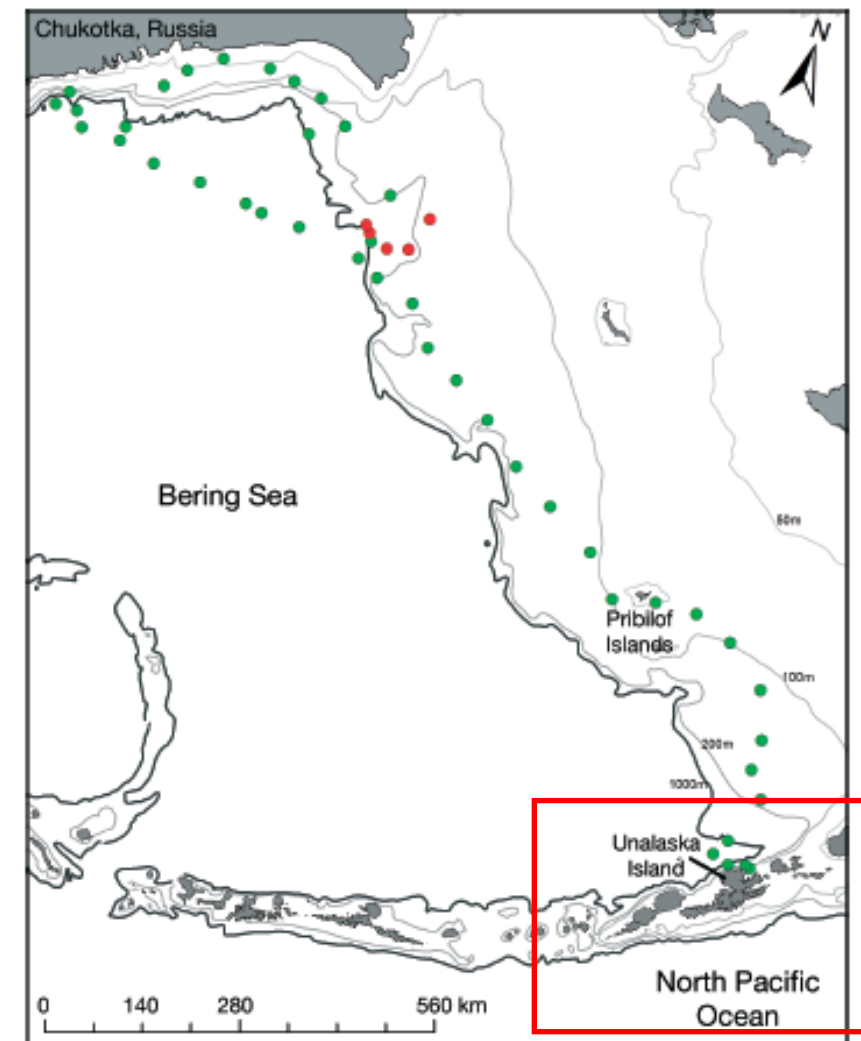
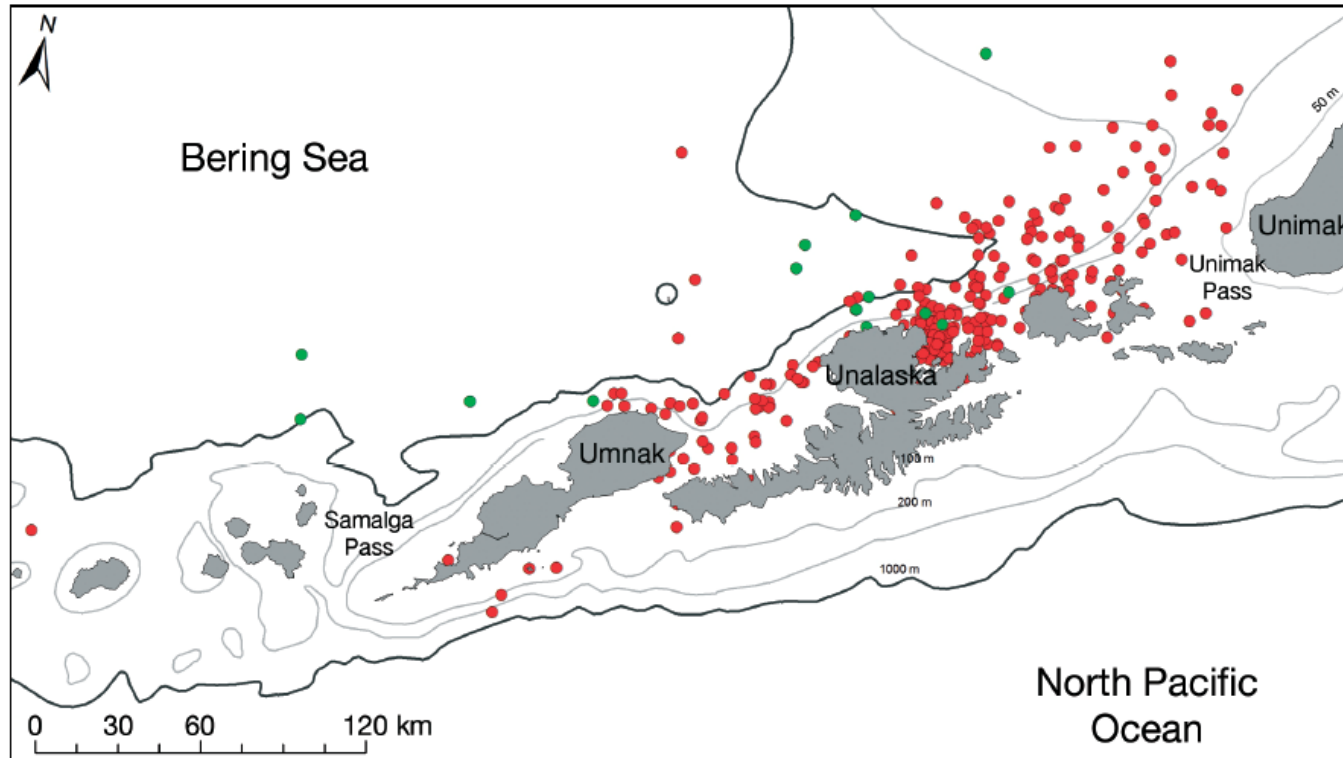
- Generate several plausible filtered/regularized tracks
- Fit an HMM to each imputed regular track
- Combine HMMs to get parameters with better uncertainty estimates



Slide from T. Michelot

HMM Examples/Applications

Humpback whales



Individual variation in movements of satellite-tracked humpback whales *Megaptera novaeangliae* in the eastern Aleutian Islands and Bering Sea

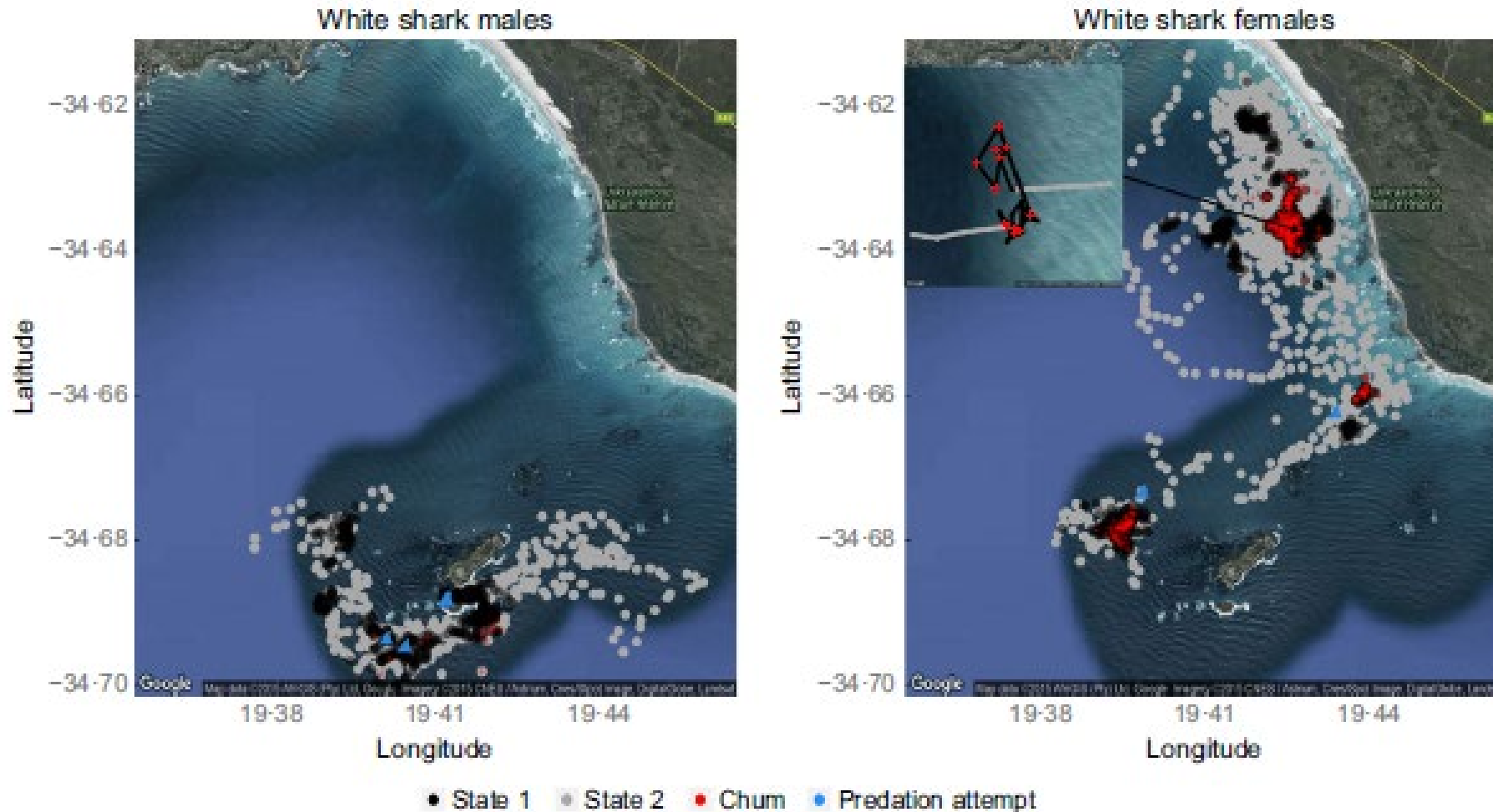
Amy S. Kennedy^{1,*}, Alexandre N. Zerbini^{1,2}, Brenda K. Rone¹, Phillip J. Clapham¹

Sex-specific and individual preferences for hunting strategies in white sharks

Alison V. Towner^{1,2}, Vianey Leos-Barajas³, Roland Langrock^{4,5}, Robert S. Schick⁴, Malcolm J. Smale^{6,7}, Tami Kaschke^{2,8}, Oliver J. D. Jewell^{2,9,10} and Yannis P. Papastamatiou¹¹

HMM Examples/Applications

White sharks



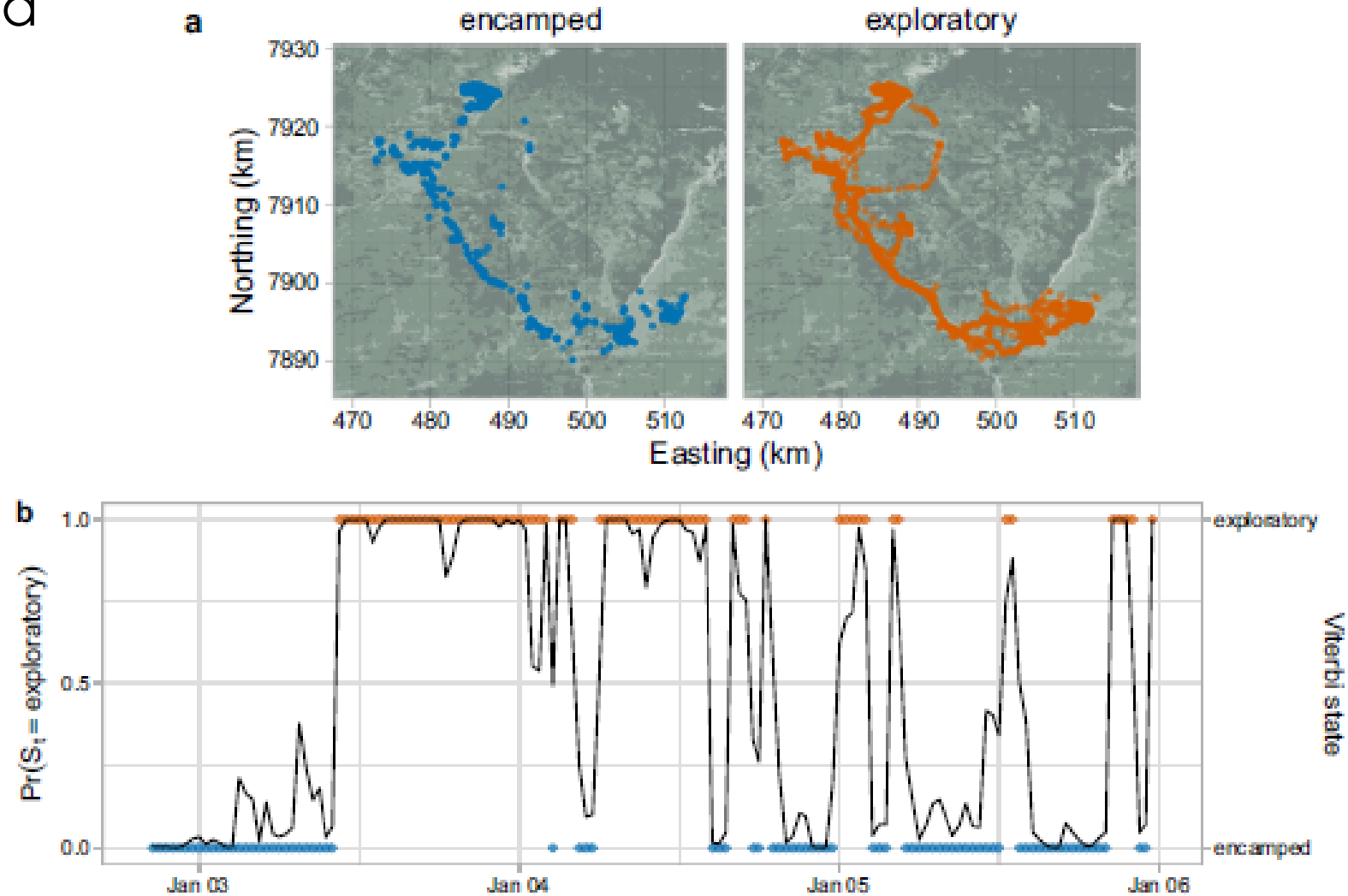
HMM Examples/Applications

Flexible hidden Markov models for behaviour-dependent habitat selection

N. J. Klappstein^{1,2*}, L. Thomas¹ and T. Michelot^{1,2}



Plains zebra



HMM Examples/Applications

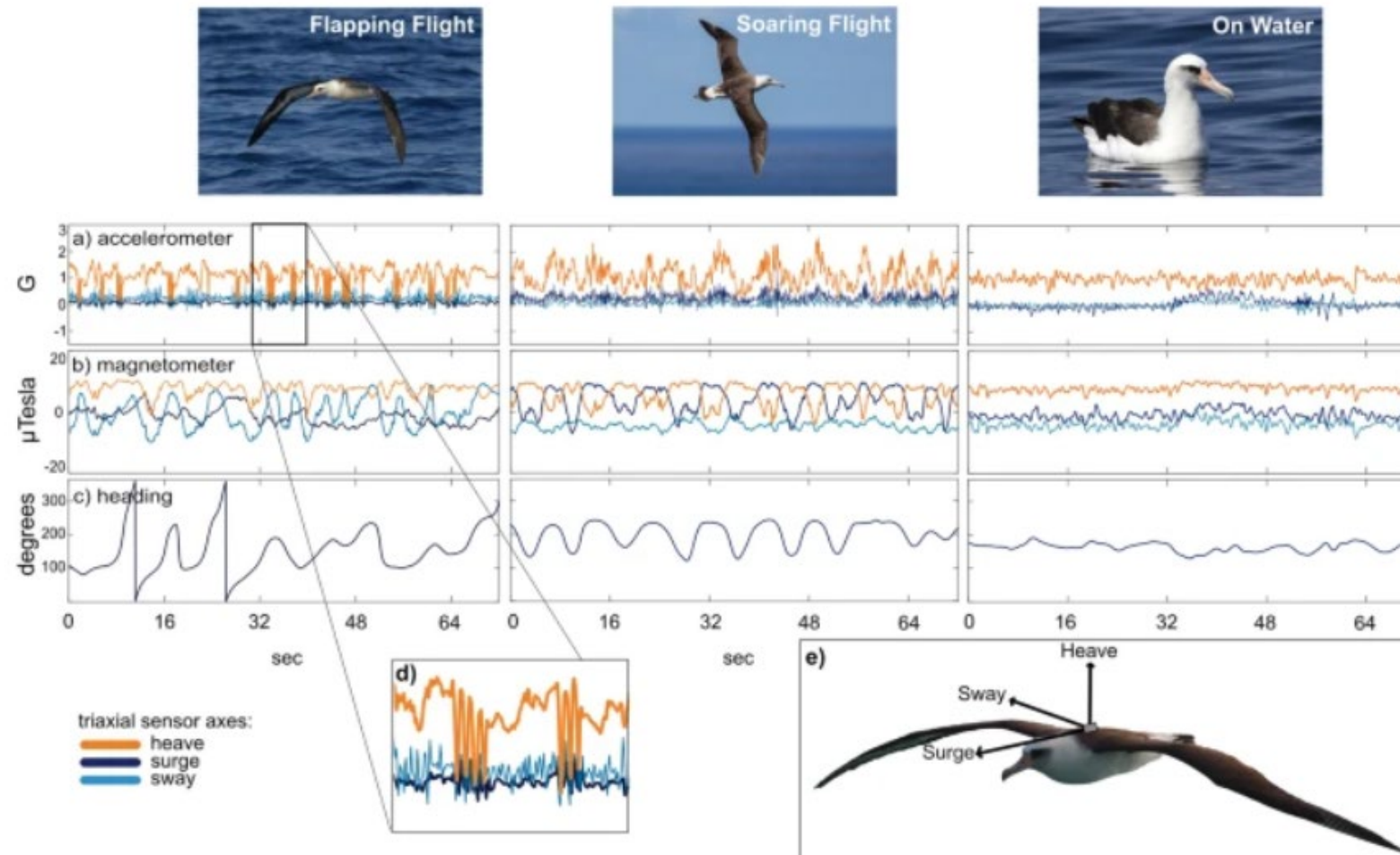
Albatross

Research | [Open access](#) | Published: 22 February 2021

Hidden Markov models identify major movement modes in accelerometer and magnetometer data from four albatross species

Melinda G. Conners [✉](#), Théo Michelot, Eleanor I. Heywood, Rachael A. Orben, Richard A. Phillips, Alexei L. Vyssotski, Scott A. Shaffer & Lesley H. Thorne

Movement Ecology 9, Article number: 7 (2021) | [Cite this article](#)



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